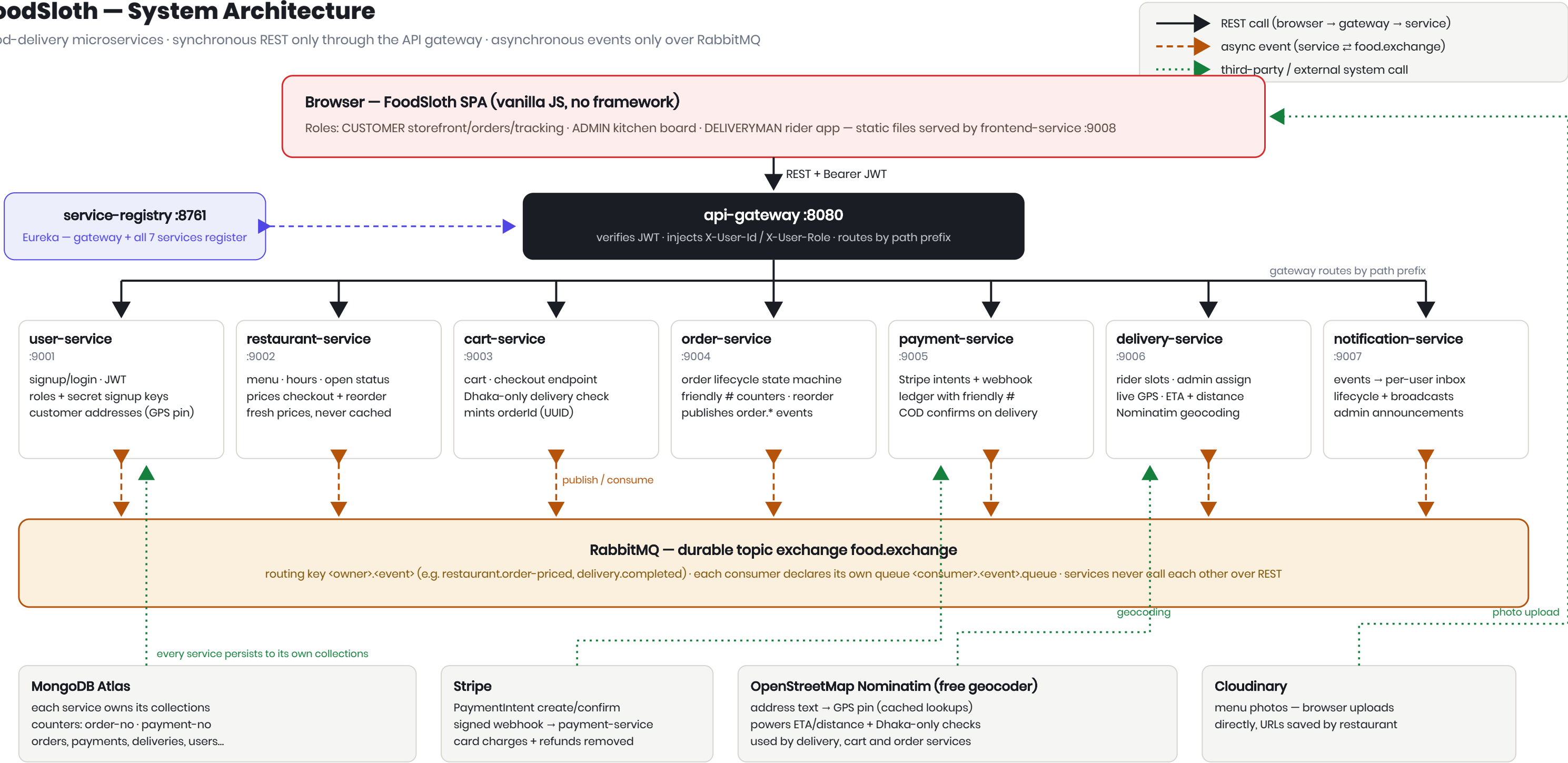


FoodSloth — System Architecture

Food-delivery microservices · synchronous REST only through the API gateway · asynchronous events only over RabbitMQ



Primary flow — checkout to delivery (solid = REST · dashed = events over food.exchange)

1. Customer checks out — cart-service enforces Dhaka-only delivery (GPS-pin box check, or geocoded address) and mints orderId → cart.checked-out.
2. restaurant-service prices the basket fresh → restaurant.order-priced (or restaurant.order-unavailable when an item is sold out).
3. order-service creates order #N (PENDING_PAYMENT) and raises payment.requested; friendly sequential numbers come from Mongo counters.
4. payment-service opens a Stripe PaymentIntent (card) or keeps the ledger PENDING (cash on delivery) → payment.succeeded / payment.failed.
5. CONFIRMED → restaurant accepts (PREPARING) → restaurant.order-ready (READY); the ready event also creates the delivery job.
6. Admin assigns a free rider (one active slot per rider); cancelling or unassigning an assignment never touches the order itself.
7. Rider app streams device GPS every 10 s over REST (POST /deliveries/riders/me/location); the customer map follows the rider live.
8. delivery.started → OUT_FOR_DELIVERY; ETA + remaining km re-computed on every GPS fix (road factor 1.3, 22 km/h, +5 min handover).
9. Addresses without a GPS pin are geocoded through Nominatim so distance/ETA still compute; (0,0) sentinel pins never enter the math.
10. Rider marks delivered → delivery.completed → order DELIVERED; the customer sees the final status on the track page and order list.
11. delivery.completed also flips the COD ledger entry PENDING → SUCCEEDED in payment-service; order.delivered feeds notification-service.
12. notification-service consumes every lifecycle event into the per-user inbox; admins can broadcast announcements to all users.
13. Reorder replays the same pipeline at today's prices via order.reorder-requested — the Dhaka restriction is enforced again.

14. Clients only ever talk REST to api-gateway; services coordinate exclusively through signed events on the shared topic exchange.

Ports — registry 8761 · gateway 8080 · user 9001 · restaurant 9002 · cart 9003 · order 9004 · payment 9005 · delivery 9006 · notification 9007 · frontend 9008

Money is stored in poisha and displayed as taka (÷100) · GPS (0,0) is the "no pin" sentinel · admin/rider signups require env-configured secret keys